The length of an SS Cygni outburst helps forecast the next: a preregistered test on a century of amateur observations

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Abstract

Dwarf novae brighten in repeated outbursts as the gas disc around a white dwarf heats up. In SS Cygni, longer outbursts tend to precede longer quiet intervals. Using a century of amateur observations from the American Association of Variable Star Observers (AAVSO), we tested whether an outburst's duration improves forecasts of when the next begins, beyond what recent quiet intervals predict. Lognormal recurrence models fitted to 1910–1959 were frozen and preregistered before the 1960–2017 observations were opened. Adding duration raised the log score by 0.0165 nats (natural-log units) per forecast (90% interval 0.0049–0.0281; 441 forecasts), meeting the preregistered criterion. The 1993–2017 half of the test period, a preregistered long/short rule and two redesigned outburst timings gave smaller gains with intervals spanning zero. None of ten other dwarf novae had pre-1990 records that qualified under frozen rules. The score rates forecasts against onsets estimated from the light curve, so linking it to physical onsets requires a validated observation model. A shared end-time estimate, or more frequent reporting while the star is active, could link duration with timing. A data-parsing error was repaired after the first data request and before any forecast was scored.

Key words: stars: dwarf novae – stars: individual: SS Cygni – accretion, accretion discs – methods: statistical

1 Introduction

Dwarf novae are close binaries in which a white dwarf accretes gas from a low-mass companion through a disc. Every few weeks to months the disc switches from a cool, faint state to a hot, bright one, and the system brightens by several magnitudes for days to weeks. The disc instability model attributes these outbursts to a thermal–viscous instability driven by hydrogen ionisation (Hameury 2020; Lasota 2001). In that picture an outburst drains part of the disc onto the white dwarf, and the next begins once the disc has refilled to a critical state.

SS Cygni (SS Cyg) is one of the brightest and best-observed dwarf novae. Amateur observers have reported its brightness to the American Association of Variable Star Observers (AAVSO) since the 1890s, producing a nearly continuous visual light curve. Its outbursts recur roughly every seven weeks and come in short and long kinds, lasting about one and two weeks (Fig. 1A).

The record has long been searched for links between successive cycles. Bath & van Paradijs (1983) related outburst energy to the following interval. Cannizzo and Mattei (1992, hereafter CM92) analysed 705 outbursts. Duration correlated with the quiet interval that followed ($r = 0.223$) and the one before ($r = 0.282$), successive durations alternated ($r = -0.400$), and durations fell into short and long classes divided at about 12 days. How a cycle is defined governs such correlations, because an onset-to-onset cycle contains the outburst's own duration (Ak et al. 2002; Cannizzo & Mattei 1992); we therefore forecast the quiet interval alone, from the end of one outburst to the start of the next. Later work extended the outburst statistics (Cannizzo & Mattei 1998), reproduced long/short alternation and slow changes in recurrence in disc calculations (Cannizzo 1993), tested for dependence between successive outburst widths (Hempelmann & Kurths 1990), and searched for low-dimensional chaos (Cannizzo & Goodings 1988) and for outburst predictors (Price et al. 2007); recurrence has also been studied in other dwarf novae (van der Woerd & van Paradijs 1987).

These studies measure association in the data they describe. A forecast test asks whether a variable improves predictions of data not used to fit the model, beyond what the model already uses. Recent quiet intervals already inform the next one, so the useful question is whether duration adds to them. The closest forecasting precedent we found tested point forecasts of onset on one held-out outburst of V363 Lyr (Grigorievskiy et al. 2014). To our knowledge, this study is the first preregistered test of probabilistic dwarf-nova recurrence forecasts (a full probability distribution for the next onset, not a single date) on held-out data, and the first held-out test of duration beyond a timing baseline.

Two explanations would make duration useful. Under the *depletion hypothesis*, a long outburst drains more of the disc, which then takes longer to refill if the companion's gas supply is similar from cycle to cycle, so longer outbursts should precede longer quiet intervals. Under the *state hypothesis*, duration tracks a slowly changing accretion state, such as the mass-transfer rate, that recent intervals describe incompletely. Separating them requires measurements beyond outburst timing (Section 4.2).

We tested the prediction with a preregistered forecasting design (Nosek et al. 2018). Forecasts were judged by a logarithmic score, which rewards the probability assigned to what happens (Gneiting & Raftery 2007), and evaluated strictly forward in time (Bürkner et al. 2020). We fitted the models to 1910–1959, froze them, published the protocol, and only then opened and scored every forecast for 1960–2017. Adding duration improved the frozen forecasts, by a margin small beside recent timing but positive through most of the test period. We then asked how the result depends on the way outbursts are timed, and whether the records of ten other dwarf novae could support the same test.

2 Data and methods

In outline, we turned nightly magnitude estimates into outburst start and end times, issued a forecast of the next start whenever an outburst had faded, and scored it against the later record.

2.1 Observations

We used visual estimates of SS Cyg from the AAVSO International Database (band `Vis.`, validation flag `V`), requested in bounded annual windows and hashed on receipt. Fainter-than reports were kept as limits. The primary model choices and fits used the development period 1910–1959. The evaluation period covers forecasts issued from 1960 January 1 to 2017 December 31 (JD 2436934.5 to 2458119.5); it contains 500,479 archive rows after deduplication, of which 249,356 meet these selection criteria. Observations from 2018 onward are reserved for a future test. Times are Julian Dates as returned by the archive.

2.2 Outburst extraction

An extraction algorithm, X4 (the fourth candidate tried), identifies outbursts as episodes brighter than visual magnitude 10, between SS Cyg's quiescent and outburst brightness. X4 groups each observer's reports into visits (reports within 0.15 day) and estimates when the light curve crossed magnitude 10 on the rise (the onset) and on the fade (the end). Each crossing time is estimated as a point and an interval from nearby reports, allowing 0.05 mag of systematic error shared across observers. For catalogue confirmation, two observers need only report magnitude 10 or brighter; a single observer needs bright reports on at least two nights, one at magnitude 9.5 or brighter, without a contradictory report at magnitude 10.5 or fainter from another observer on those nights. Duration runs from fitted onset to fitted end, and a quiet interval from one outburst's end to the next onset. A quiet interval counts as covered when no gap between visits within it exceeds seven days. X4 was adopted during development after three earlier extractors failed a synthetic recovery screen, a test of recovering known outbursts from simulated light curves; X4 passed it at magnitude 10 (Appendix A).

2.3 Forecasts

A separate causal rule issues forecasts: after magnitude-9.5-or-brighter reports from two observers, or one observer on two nights, subsequent reports at magnitude 10.5 or fainter from two observers confirm the fade. Every such confirmation issues a forecast, including repeats. Each forecast uses only reports dated within the preceding 400 days: X4 is rerun on that window alone, and tests that insert later rows confirm that no predictor changes.

Each forecast is a lognormal distribution for the quiet interval Q that follows the fitted end e of the completed outburst. With log-location μ and log-scale σ, the three models (numbered as in the protocol) are

$$\begin{aligned} \text{M0}: &\ \mu = \alpha, \\ \text{M2}: &\ \mu = \alpha + \gamma_1 q + \gamma_2 L, \\ \text{M3}: &\ \mu = \alpha + \gamma_1 q + \gamma_2 L + \beta x, \quad \beta \geq 0, \end{aligned}$$

where q is the log of the previous covered quiet interval, L the mean of up to six earlier covered log quiet intervals ($L = 0$ when none is available) and x the log duration, each standardised with development means and standard deviations. M2 is the timing baseline and M3 adds duration; the constraint $\beta \geq 0$ restricts duration's effect to the direction CM92 reported, longer outbursts before longer quiet intervals. A forecast

issued at age $a = t_{\text{issue}} - e$, the time already elapsed since the fitted end, is the lognormal conditioned on $Q > a$. Forecasts without a covered previous interval use M0 under both models, and forecasts without a fitted end use one shared forecast of the remaining wait. These fallback forecasts contribute exactly zero to the comparison. Table 1 summarises the models and test; Fig. 1B shows one cycle schematically.

2.4 Outcomes and the working score

The outcome is the onset of the next outburst as X4 estimates it from the later record, an interval $(l, u]$. A forecast is scored until the first of three events: loss of coverage (a visit gap over seven days), the next issued forecast, or the end of the evaluation period. If no onset is established by then, the forecast is right-censored at that time c and scored on the probability of no onset before c. The *working score* of a forecast is

$$s = \begin{cases} \log\left[S(l-e) - S(u-e)\right] - \log S(a) & \text{onset interval,} \\ \log S(c-e) - \log S(a) & \text{censored,} \end{cases}$$

where S is the lognormal survival function. The score is the log probability, in nats, that the forecast assigned to the extracted onset window, or to no onset before c; higher is better. We call it the working score because it rates extracted windows rather than physical onsets. The paired gain is M3's score minus M2's for the same forecast. Exponentiating a mean gain gives the ratio of the geometric-mean probabilities the two models assigned to the scored outcomes.

2.5 Fitting, freezing and the test

M2 was fitted by maximum likelihood to the 307 timing-eligible development forecasts and frozen. The duration coefficient was then fitted with M2 held fixed, so that $\beta = 0$ reproduces M2 exactly. The frozen values are $\beta = 0.0466$ per standard deviation of log duration, $\gamma_1 = 0.0780$, $\gamma_2 = 0.0513$ and, shared by M2 and M3, $\sigma = 0.346$.

The single preregistered statistic is the mean paired gain over all scored forecasts issued in 1960–2017, including the zero contributions of fallback forecasts. Because forecasts close in time are not independent, its standard error is cluster-robust over consecutive six-year blocks (2191.5 days) from the start of 1960, with a $G/(G-1)$ correction for the G nonempty blocks and Student's t with $G-1$ degrees of freedom (Cameron & Miller 2015). The criterion was a one-sided 95% lower bound above zero, equivalent to a two-sided 90% interval wholly above zero. Two secondary comparisons were also specified: an era split at 1993, after the end of CM92's record, and a rule that replaces continuous duration with CM92's 12-day long/short classes plus within-class duration. Before the development fit, the criterion was met in 11 of 40 simulated evaluations with an effect sized to CM92's correlations, and in 3 of 220 simulated histories with no effect.

2.6 Access and execution

The protocol was deposited on Zenodo (doi:10.5281/zenodo.24053338) on 2026 November 8, before any evaluation data were requested (Fig. 1C). The first request stopped before any forecast was scored, because the archive returned the literal token `None` for missing uncertainties, a field no analysis uses. The parser was changed to accept that exact token without altering or dropping a row. Independent review, invariance tests and a full synthetic rehearsal passed before the run resumed; forecasting code, parameters and protocol files stayed unchanged. Two independent implementations of the forecast table, which records every forecast and its inputs, had to agree exactly before scoring, and a separate checker reproduced every score.

2.7 Measurement checks after the result

An external review of the result asked how far it depends on the way outbursts are timed. Each analysis below was specified before its own fits, and all reuse the opened data.

- **Rescoring** audited the censoring rules, scored each forecast against X4's point estimate of onset instead of its interval, and added synthetic errors to fitted end times.
- **Report windows and clocks** rebuilt each onset outcome as a report window, from the last faint report to the first decisively bright one (a second extractor, X5), and compared two clocks, that is,

two ways of starting the quiet interval. The fitted-end clock keeps the fitted end. The decisive-faint clock starts at the first decisively faint report and measures duration as the span of decisively bright reports, so no end time is interpolated. These comparisons use new windowed timing baselines: B0 with recent-interval covariates, B1 with discounted history, and B2, a state-space level model chosen in advance.

- **Drift recalibration** recentred duration on the preceding ten outbursts and updated the baseline level from preceding outcomes, to ask how slow change in the typical quiet interval (drift) affects the gain.

2.8 Transfer screen

To ask whether other dwarf novae could support the same test, we froze the screen's code in Git before requesting any of their data. Eight candidates (SS Aur, RU Peg, AR And, X Leo, TW Vir, UU Aql, CH UMa and DX And) and two eclipsing systems held to stricter rules (U Gem and IP Peg) were chosen from catalogue properties, with thresholds proposed from the same catalogues. Only records before 1990 (JD $<$ 2447892.5) were opened, keeping later decades for a future test. A star qualified if its pre-1990 record yielded at least 40 supported forecasts (with a covered previous interval, a measurable duration and a scorable outcome), at least 70% eligible onset windows and stable timing-baseline fits. It also needed no operational long-run flag, which marks bright runs longer than ordinary outbursts (60 days or more, or repeated 40-day runs); the eclipsing systems also had to pass an eclipse-handling check. The screen fitted no duration effect and computed no forecast gain (Appendix E).

Table 1. Models and test.

Item	Specification
Forecast	Lognormal quiet interval after the fitted end, conditioned on survival to issue
M0	Constant log-location
M2, timing baseline	Log previous covered interval; mean of up to six earlier covered log intervals
M3	M2 plus log duration, $\beta \geq 0$, fitted with M2 fixed
Development fit	1910–1959; frozen $\beta = 0.0466$, $\gamma_1 = 0.0780$, $\gamma_2 = 0.0513$; M2/M3 $\sigma = 0.346$
Evaluation	Every forecast issued 1960–2017
Working score	Log probability of the extracted onset window, or of survival to censoring, given survival to issue
Statistic and criterion	Mean M3 − M2 gain; six-year cluster standard error; one-sided 95% lower bound above zero
Redesigned comparisons	Report-window outcomes; fitted-end and decisive-faint clocks; baselines B0, B1, B2

3 Results

3.1 Duration improved the frozen forecasts

Over 441 forecasts issued in 1960–2017, adding duration raised the mean working score by 0.0165 nats (90% interval 0.0049 to 0.0281; ten blocks, nine degrees of freedom), meeting the preregistered criterion (Fig. 2). The duration model forecast longer waits after longer outbursts: with timing history fixed, doubling the duration lengthens the median forecast quiet interval, before conditioning on the time already elapsed, by about 9%. The geometric-mean probability assigned to the scored outcomes rose by 1.7%. The timing baseline M2 itself gained 0.117 nats over the constant forecast M0, so duration added about 14% of what recent timing adds. The outcomes comprised 435 onset intervals and six censored forecasts; five of the forecasts used the shared remaining-wait fallback.

Fig. 2A shows the first eligible forecast of the evaluation period, chosen by date. It followed a shorter-than-average outburst, so M3 gave a median remaining wait of 26.77 days against 27.57 for M2, and placed 0.603 of its probability within 30 days against 0.577. The gain was positive in eight of ten six-year blocks (Fig. 2B),

and leaving out any one block gave means from 0.0132 to 0.0194. Preregistered variants (uniform end times, a previous-gap-only baseline, a two-day issue delay and three-year clusters) each kept the lower bound above zero (Appendix B). M3's predictive intervals were conservative, wider than nominal: nominal 50% intervals covered between 0.546 and 0.617 of outcomes, and nominal 90% intervals between 0.923 and 0.980.

The preregistered era split gave 0.0240 nats (0.0091 to 0.0389; 261 forecasts) for 1960–1992 and 0.0057 (−0.0107 to 0.0221; 180 forecasts) for 1993–2017. The long/short rule gained 0.0025 nats (−0.0022 to 0.0072).

3.2 Measurement checks

The gain survived rescoring against point onsets and shrank under report-window outcomes (Fig. 3). Scoring against X4's point estimates of onset gave 0.0154 nats (0.0034 to 0.0274), and the censoring audit left all 441 outcomes unchanged. The report-window comparisons gave smaller gains with intervals spanning zero: 0.0075 nats (−0.0008 to 0.0159; 432 forecasts) on the fitted-end clock and 0.0043 nats (−0.0008 to 0.0094; 431 forecasts) on the decisive-faint clock, both with baseline B2. The paired contrast between the decisive-faint and fitted-end gains at the same issue times was −0.0031 nats (−0.0078 to 0.0017; Appendix C).

To ask whether end-time errors alone could create a gain, we added independent errors to fitted end times in synthetic null and gap-dependent histories; mean gains were zero or below at every tested error size. To gauge how much forecast information observing effort carries, we replaced duration with the observer-night reporting rate during the outburst: on the decisive-faint clock it gained 0.0030 nats (0.0013 to 0.0048) over baseline B2. Separately, adding the end-cadence gap (from the last report short of decisive faintness to the decisively faint report that starts the clock) to that baseline left the duration gain unchanged at 0.0043 nats.

Of the redesigned baselines, B1 fitted the development data best and B2, chosen in advance for the main report-window comparisons, worst (negative log likelihoods 827.24 for B1, 830.19 for B0 and 849.28 for B2). The typical quiet interval early in the evaluation period was shorter than in the development blocks. Recentring duration on the preceding ten outbursts and recalibrating the baseline level from preceding outcomes reduced the duration gain in the redesigned comparison closest to the primary design (fitted-end clock, baseline B0) from 0.0184 to 0.0132 nats, and the reporting-rate gain from 0.0030 to 0.0014. After this recalibration the duration gains stayed positive on every clock and baseline, from 0.0052 to 0.0132 nats, and the fitted-end B0 gain kept a positive lower bound.

3.3 Transfer screen

None of the ten stars qualified (Fig. 4; Appendix E). Nine yielded fewer than the 40 supported forecasts required, and across all ten the frozen pre-1990 records yielded 217. SS Aur yielded 110, with 79.6% eligible onset windows, and failed the long-run check with a 100.97-day bright run made up of four separate clusters of report nights. The eclipsing systems U Gem and IP Peg yielded 27 and 0 supported forecasts, and neither passed the preliminary stage of the eclipse check.

The candidates' records supported far fewer forecasts per outburst than SS Cyg's. In 1980–1989 SS Cyg's record yielded 67 supported forecasts against 73 confirmed outbursts; the ten candidates together yielded 102 against 485 (Fig. 4B). The largest single loss was the requirement for a covered previous quiet interval: in a separate count from the raw reports, 822 pre-1990 candidate forecasts with a clock fell to 280 once that requirement applied. At the 1980s rate, all ten stars together, ignoring the other qualification rules, would yield 367.2 supported forecasts over 36 years, well short of the 1,100 assumed in planning a multi-star test (Appendix F).

4 Discussion

4.1 What the result shows

Adding the length of the last outburst to recent quiet intervals improved forecasts of SS Cyg's next onset, in a test whose models and criterion were fixed before the evaluation data were opened. The direction matches CM92's correlation between duration and the following quiet interval, now carried into forecasts of 58 years of later observations. The gain is modest, about a seventh of the gain from timing history, and positive in eight of ten six-year blocks.

Rescoring against point onsets gives a similar gain, and report-window outcomes give smaller gains whose intervals span zero. Used in place of duration, the reporting rate carries less forecast information, and adjusting the baseline for the reporting gap at the fade leaves the duration gain unchanged.

4.2 Depletion or state?

Both hypotheses accommodate the observed direction. The depletion hypothesis ties the effect to individual cycles: each long outburst should lengthen the interval that follows. Disc models that drain different fractions of the disc in long and short outbursts reproduce the observed alternation (Cannizzo 1993), and alternation is strong in these data: the correlation between successive log durations was -0.52 in the evaluation years. In exploratory fits that re-estimate the level on the evaluation data, a long/short indicator had a higher likelihood than continuous duration (Appendix D), although the frozen long/short rule gained little (Section 3.1). This motivates an outburst-type predictor in a future frozen comparison. Type-dependent refilling and a changing state could each produce such a long/short pattern.

The state hypothesis ties the effect to slower change: duration would carry information about the mass-transfer rate or disc state that the recent-interval average smooths over. SS Cyg's behaviour can depart from its usual pattern; an anomalous event was reported in 2021, after the evaluation period (Kimura et al. 2021). The drift in the typical quiet interval between development and evaluation, and the smaller gains after duration recentring and baseline recalibration, are compatible with slow variation, in the star or in how it was observed.

Forecasts that add an independent measure of accretion state, such as quiescent brightness (Vogt & Vega-Manubens 2023) or X-ray and extreme-ultraviolet flux (Wheatley et al. 2003), could separate the two: if duration's gain shrinks once state is known, the state hypothesis gains support; if it persists, depletion does. The frozen forecasts can also be scored on SS Cyg's observations from 2018 onward, which remain unopened.

4.3 What a record needs

SS Cyg's record supplies enough successive, well-covered cycles for the frozen test. A forecast built on the previous interval needs uninterrupted coverage across two cycles, and one observing gap of more than a week removes it. Under the same rules the candidates' pre-1990 records yielded about one supported forecast per five confirmed outbursts, against nearly one per outburst for SS Cyg. Their records from 1990 onward, and automated surveys, may supply more supported cycles; a multi-star test would need its own frozen design on those records.

The design itself is general. Fitting on early records, freezing and publishing the forecasts, and scoring them on later records suits other long, irregular records of recurrent events, such as volcanic eruptions or disease outbreaks. It needs enough well-covered successive events, a baseline that captures their recent rhythm, and a score fixed before the test data are seen.

5 Limitations

Table 2 lists the assumptions behind the primary result, what tested each and what remains open. Three matter most.

First, the working score evaluates extracted onset windows, which are fitted statistical summaries. Establishing skill at forecasting physical onsets requires a validated model of how the selected reports and observing times arise. It must show that reports are assigned to the right outbursts and forecasts to a valid risk set, that each window contains the true crossing, and that, given the retained history and forecast state, the complete reporting and selection process gives the same probability of obtaining the reported window for every onset time it contains, with observation parameters distinct from forecast parameters. Under these conditions, known as coarsening at random (Heitjan & Rubin 1991), the observation term factors out of the likelihood (Appendix C). X4 passed its recovery screen on light-curve shapes known during development; recovery on a family of shapes held out from development remains untested, because that benchmark failed qualification before X4 was scored on it.

Second, duration and the following quiet interval share one estimated end time, so errors in that time can couple them, and observers may report more often when the star is active. The decisive-faint clock removes interpolation from the end but keeps report timing, and a constructed example shows that report timing alone can produce a positive duration gain. The endpoint-error experiments added independent Gaussian errors in a fixed simulator, and the reporting rate itself carried forecast information.

Third, all post-result SS Cyg reanalyses reuse opened data and are exploratory. The redesigned comparisons and the drift recalibration change baseline, clock, coding and scored population together, so their smaller gains cannot be read as fractions of the primary gain. A richer timing baseline could also absorb part of that gain.

The evaluation followed a parser repair made after the first data access and before scoring; the protocol DOI timestamps the original code, and the repair is documented separately. Forecasts use reports by observation date; when historical reports were uploaded or revised is unknown, so the test replays the archive as it stands today. The transfer screen's ratios count forecasts and outbursts by different landmarks and describe yield under its frozen rules, which used catalogue thresholds. The result concerns one star.

Table 2. Assumptions behind the primary result.

Assumption	Why it matters	What tested it	What remains open
Extracted onset windows stand for physical onsets	Defines what the score rewards	Synthetic recovery screen; point rescoring; report windows	A validated observation model; recovery on held-out light-curve shapes
Endpoint measurement model	A shared estimated end couples duration and the next interval	Sign constraint; simulated independent end errors; decisive-faint clock	Systematic or state-dependent errors; report-timing coupling
Observation process is coarsened at random given history	Observers may watch an active star more closely	Reporting-rate mark; end-cadence covariate	The rate itself carries information
The timing baseline is adequate	The gain is measured beyond it	Baselines B0–B2; drift recalibration	Richer baselines; recalibration is exploratory
The archive reflects what was reported at each date	Defines the information at issue	Prefix-only predictors; inserted-row tests	Upload and revision dates are unknown
Six-year blocks are independent	Sets the interval width	Leave-one-block-out; three-year clusters	A standard error 1.43 times larger removes the result
The evaluation followed the protocol	Confirmatory status	DOI before access; two implementations; independent checker	Parser repair after first access
One star	Scope	Ten-star transfer screen	No other star qualified

6 Conclusions

Knowing how long an SS Cygni outburst lasted improves forecasts of when the next one begins. Forecasts that included duration, fitted to 1910–1959 and frozen before 1960–2017 was opened, assigned more probability to the extracted onsets than forecasts from recent timing alone; the gain was small and met a preregistered criterion. A century of amateur observations made the test possible. Separating disc depletion from a slowly changing accretion state, and repeating the test on other dwarf novae, will need records with enough well-covered successive cycles, together with independent measures of accretion state. The frozen forecasts, code and protocol are public, ready to be scored on SS Cyg's observations from 2018 onward.

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[Placeholder: generative-AI declaration, to be written before submission. It will state the roles of GPT-6 Pro, Claude Opus 5.5 and GPT-6 Codex, and how their calculations, analyses and figures were tested and validated, as the journal requires.]

Data availability

Protocols, source code, frozen parameters, aggregate evidence, the claims ledger, figures and access receipts are at https://github.com/BWBrook/dwarf-nova-memory (code MIT; text, figures and derived data CC BY-SA 4.0). The evaluation protocol is archived at https://doi.org/10.5281/zenodo.24053338. AAVSO observations are available from the International Database. Raw extracts, observer identities, event catalogues and per-forecast scores are kept outside the repository; reproduction uses their hashes without a new archive request.

Software

Analyses used Python with NumPy (Harris et al. 2020) and SciPy (Virtanen et al. 2020); figures used Matplotlib (Hunter 2007). The manuscript was built with Pandoc and makedoc (https://github.com/BWBrook/makedoc).

Appendix A: Extraction, issuance and access

Extraction. X4 (`src/dnmem/extract_x4.py`) forms its crossing summaries from normalised slope-mixture fits to nearby estimates. The 0.05-mag shared allowance stops the intervals shrinking without limit as estimates accumulate; it was chosen on development simulations. The issuing rule arms only on bright evidence at least 0.5 mag past the threshold. Three earlier extractors, X1–X3, failed qualification. X4 passed a synthetic screen at magnitude 10 on the qualified cadence family, where the seven-day coverage rule was also chosen; magnitude-9.5 diagnostics failed in later eras.

Issuance and predictors. The current episode is the last episode in the 400-day window whose end bracket reaches the issuing bright evidence. A duration of zero or less makes the mark unavailable. The previous interval counts only if the previous episode begins more than 30 days inside the 400-day window. A two-day reporting delay was a frozen sensitivity.

The sign constraint. Shared end-time error pushed fitted null duration coefficients negative in development simulations, which motivated fitting β with M2 fixed and bounding it at zero. The bound restricts direction; an exact construction with constant true duration can still yield positive information, so endpoint coupling remains a stated limit (Section 5).

Access timeline (2026 November 8, UTC). Protocol published 12:34:47; first request 12:38:28, stopped 12:39:51 before any ledger or score; repaired acquisition 12:51:18; ledger concordance and primary result 13:06:58. The repair freeze is commit `5e39aeb` and the execution commit `d7151e0`. No person or agent inspected real scores, outcomes, counts or coverage summaries before the primary result was written. A separate checker reproduced all 2,641 variant score rows to order 10^{-13}.

Appendix B: Frozen sensitivities and calibration

Table B1. Frozen comparisons: mean paired working-score gain (nats per forecast) and 90% cluster interval. Only the first row is confirmatory.

Comparison	Forecasts	Gain	Interval
Primary, M3 − M2, 1960–2017	441	0.0165	0.0049 to 0.0281
1960–1992	261	0.0240	0.0091 to 0.0389

Comparison	Forecasts	Gain	Interval
1993–2017	180	0.0057	−0.0107 to 0.0221
12-day class and within-class rule	441	0.0025	−0.0022 to 0.0072
Uniform-end marks	441	0.0166	lower 0.0044
Previous-gap-only baseline	441	0.0170	lower 0.0054
Two-day issue delay	441	0.0165	lower 0.0049
Joint-valid forecasts	436	0.0167	lower 0.0049
Three-year clusters	441	as primary	lower 0.0059
Baseline M2 − M0	441	0.1168	0.0663 to 0.1672

Deleting the six censored forecasts gives 0.01678 nats. Coverage bounds for M3's central intervals (Fig. 5) run from definitely to possibly covered, given the extracted windows and censoring; they describe this sample. The frozen placebo with a lagged duration equals M2, because its development coefficient sat at the bound of zero.

Appendix C: Report windows, clocks and baselines

Report windows. X5 searches for decisive faintness after the previous peak before selecting the next decisive-bright rise, and keeps raw time supports and flags. It yielded 818 available and 805 eligible windows among 845 confirmed episodes, with median width 1.15 days. The original report clips were missing for 434 of the original associations, so the earlier report-interval rescoring covered only 11 contrasts.

Premises. Treating a report window as the onset outcome rests on three premises: correct association of reports with outbursts, completeness and a valid risk set (A0); containment of the true crossing in the window, or genuine survival through a censoring time (A1); and a complete, normalised observation process that is coarsened at random (A2). Under them the report likelihood factors into an observation term and the forecast probability of the window. Establishing them needs evidence beyond visit timing and margins past threshold.

Clocks. The decisive-faint clock anchors each forecast at the first decisively faint report, uses the decisive-bright span as duration, and waits for report availability before issuing. Exact coincident issues share one representative, and unsupported outcomes stay unsupported.

Table C1. Report-window contrasts. H3 is the paired difference between the H2 and matched H1 gains.

Exploratory contrast	Scored issuances	Mean gain, nats	Six-year 90% interval	Twelve-year 90% interval
H1: fitted-end clock, B2 duration	432	0.0075	−0.0008 to 0.0159	−0.0014 to 0.0165
H2: observed clock, B2 bright span	431	0.0043	−0.0008 to 0.0094	−0.0021 to 0.0107
H3: observed-clock gain minus matched fitted-end gain	431	−0.0031	−0.0078 to 0.0017	−0.0078 to 0.0017

Baselines and drift. The drift recalibration subtracted the mean of the preceding ten marks and recalibrated the baseline level from preceding evaluation outcomes, keeping the development coefficient. It used

final saved labels at proxy availability times, and it is a diagnostic of the historical record, distinct from a prospective forecasting algorithm.

Appendix D: Long/short classes

With M2's historical parameters fixed and a free common level, likelihood-ratio statistics on the evaluation data were 14.7 for the class indicator against level only, 11.7 for continuous log duration against level only, and 0.17 for adding within-class duration to the class indicator. The class contrast corresponds to a 14% larger working clock scale after long outbursts (Fig. 6). The log class offset was 0.16323 (statistic 13.504) in 1960–1992 and 0.07828 (2.262) in 1993–2017. Giving both frozen rules M2's evaluation-fitted level raised the class rule's gain to 0.00905 nats (0.00409 to 0.01401). A long outburst followed a long one in 0.1495 of forecast-record pairs and a short one in 0.6640. These fits reuse the evaluation outcomes.

Appendix E: Transfer screen

The screen's code was fixed in Git at commit `d827313` before any candidate record was requested. Names were resolved under the same date restriction, and pagination, identity, row-window and duplicate checks preceded analysis. Supported forecasts used the decisive-faint clock and needed a positive mark and an eligible onset window or positive-exposure censoring. Baseline stability was tested on B0 fits without marks. The long-run rule flags a supported bright run of at least 60 days, or repeated 40-day runs, with minimum night and observer support. The eclipsing systems used two derived observer-night streams, which had to agree, and a synthetic eclipse test whose no-eclipse baseline had to recover enough events first; neither system passed that baseline.

Table E1. Frozen qualification. Eclipsing-system counts use the median derived-night stream.

Candidate	Supported development issuances	Eligible onset windows	Main failed conditions
SS Aur	110	79.6%	Operational morphology
RU Peg	23	63.5%	Count, windows
AR And	7	31.4%	Count, windows, base information/fit
X Leo	35	44.0%	Count, windows
TW Vir	0	11.6%	Count, windows, base information/fit
UU Aql	15	80.8%	Count
CH UMa	0	70.4%	Count, base information/fit
DX And	0	70.8%	Count, base information/fit
U Gem, hold	27	90.6%	Count, eclipse guard inconclusive
IP Peg, hold	0	66.7%	Count, windows, base information/fit, eclipse guard inconclusive

Attrition. In the ordinary report-level diagnostic, 822 clock-available pre-1990 forecasts fell to 280 with a covered previous interval and 274 after requiring a mark. Observing holes, ineligible onset windows and association all contributed to the missing history.

SS Aur. Separate parsing found 252 positive visit medians in the flagged run, of which 176 lay in the band $12.5 < m \leq 13.5$, just fainter than the decisive threshold. Its decisive-bright nights form four groups of one, seven, three and five nights. Removing this flag would leave SS Aur the only star above the count gate.

The eligible-window fraction and supported count of every star are shown in Fig. 7.

Appendix F: Precision planning

Synthetic studies asked how many stars and forecasts a multi-star test would need. They simulated development fits, guarded marks, delayed labels and rolling level updates in Gaussian log gaps, with a nominal per-star gain of 0.004 nats before mark noise and learning. Two statistics answer different questions. For a unit-variance Gaussian model with standardised mark x, true coefficient β and frozen forecast shift bx,

$$E[U_0] = \beta, \qquad E[\Delta s] = b\beta - b^2/2,$$

so a positive score moment U_0 can accompany a negative gain Δs by the actual forecast; with $\beta = 0.05$ and $b = 0.20$, the expected gain is -0.010 nats.

Table F1. Whole-star rejection frequencies at a nominal per-star gain of 0.004 nats.

Stars and condition	Nominal evaluation issuances	Score-moment rejection	Frozen-gain rejection
Eight, clean	1,100	81.8%	36.7%
Eight, persistent / small development / missing marks	1,100	63.6%	—
Eight, noisy / delayed / common state	1,100	38.7%	—
Six, clean	1,100	76.8%	33.1%
Six, clean	2,000	94.6%	48.8%
Six, persistent / small development / missing marks	1,100	59.4%	21.3%
Six, noisy / delayed / common state	1,100	36.2%	10.3%

The grids used 70,000 and 16,000 trials. Under their stressed conditions, both leave the required sample size open.

Appendix G: Prior art

CM92's Table 4 gives, for 705 outbursts from 1896 to 1992, correlations of duration with the following onset-to-onset cycle of 0.475 and with the preceding cycle of 0.145; for end-to-end cycles the pattern reverses (0.093 and 0.521). An onset-to-onset cycle contains its first outburst's duration, so this study's target is the quiet interval alone. Durations split at 12 days into 401 short and 304 long outbursts. A bounded independent search found no earlier preregistered probabilistic held-out recurrence evaluation; its queries and receipts are in `data/provenance/a10m_novelty/`.

References

Ak, T., Ozkan, M. T., & Mattei, J. A. 2002, Astronomy & Astrophysics, 389, 478

Bath, G. T., & van Paradijs, J. 1983, Nature, 305, 33

Bürkner, P.-C., Gabry, J., & Vehtari, A. 2020, Journal of Statistical Computation and Simulation, 90, 2499

Cameron, A. C., & Miller, D. L. 2015, Journal of Human Resources, 50, 317

Cannizzo, J. K. 1993, The Astrophysical Journal, 419, 318

Cannizzo, J. K., & Goodings, D. A. 1988, The Astrophysical Journal, 334, L31

Cannizzo, J. K., & Mattei, J. A. 1992, The Astrophysical Journal, 401, 642

Cannizzo, J. K., & Mattei, J. A. 1998, The Astrophysical Journal, 505, 344

Gneiting, T., & Raftery, A. E. 2007, Journal of the American Statistical Association, 102, 359

Grigorievskiy, A., Mantere, M., Akusok, A., Eirola, E., & Lendasse, A. 2014, in International Work Conference on Time Series Analysis (ITISE 2014), Vol. 1, 520, https://research.cs.aalto.fi/aml/Publications/Publication216.pdf

Hameury, J. M. 2020, Advances in Space Research, 66, 1004

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357

Heitjan, D. F., & Rubin, D. B. 1991, The Annals of Statistics, 19, 2244

Hempelmann, A., & Kurths, J. 1990, Astronomy and Astrophysics, 232, 356

Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90

Kimura, M., Yamada, S., Nakaniwa, N., et al. 2021, Publications of the Astronomical Society of Japan, 73, 1262

Lasota, J.-P. 2001, New Astronomy Reviews, 45, 449

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. 2018, Proceedings of the National Academy of Sciences, 115, 2600

Price, A., Henden, A. A., Foster, G., et al. 2007, Publications of the Astronomical Society of the Pacific, 119, 1361

van der Woerd, H., & van Paradijs, J. 1987, Monthly Notices of the Royal Astronomical Society, 224, 271

Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, Nature Methods, 17, 261

Vogt, N., & Vega-Manubens, I. 2023, Monthly Notices of the Royal Astronomical Society, 525, 3682

Wheatley, P. J., Mauche, C. W., & Mattei, J. A. 2003, Monthly Notices of the Royal Astronomical Society, 345, 49

424 **Fig. 1.**

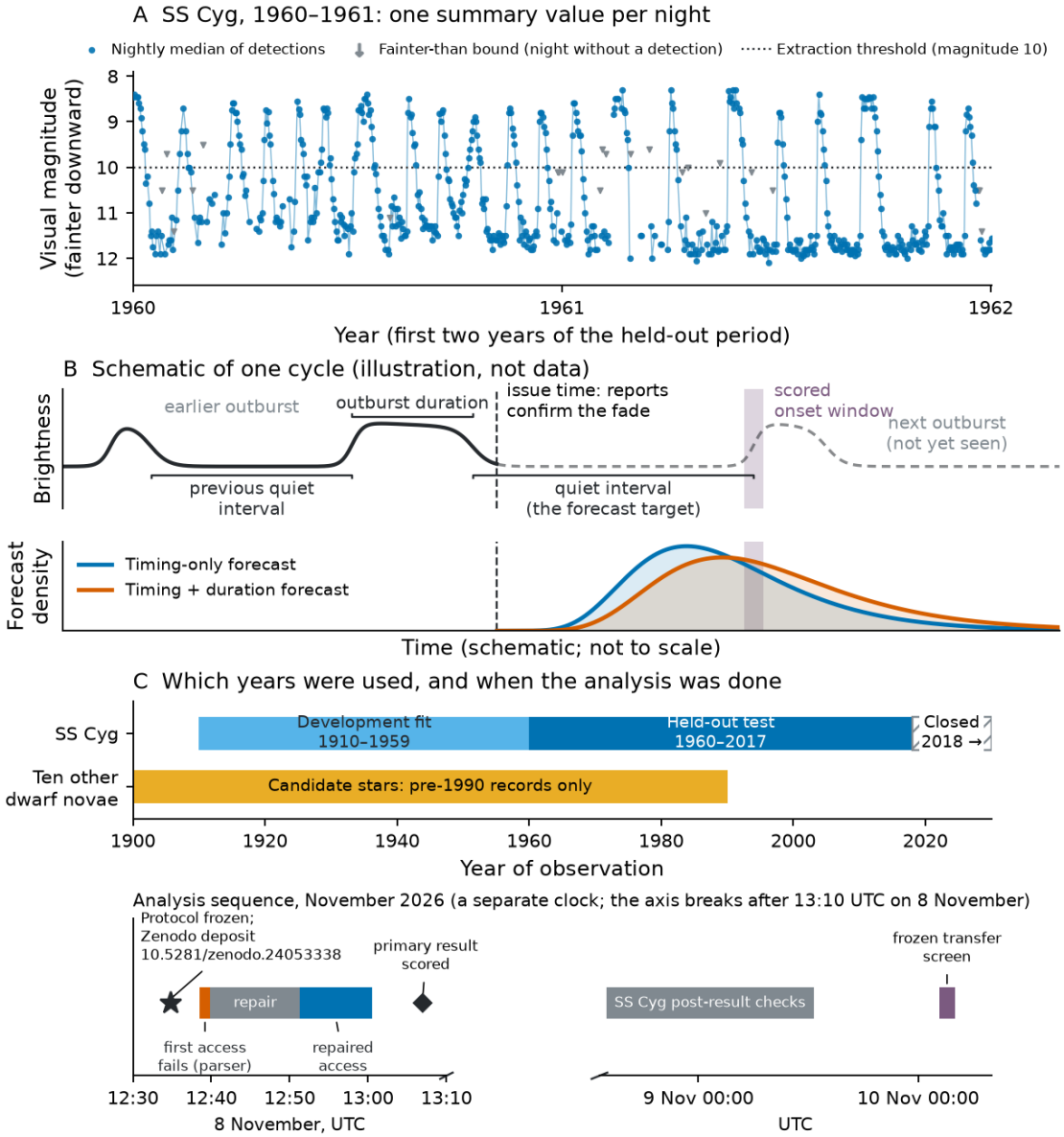


Figure 1: SS Cyg and the test design. (A) SS Cyg in 1960–1961, the first two held-out years: nightly medians of detections, and fainter-than bounds on nights without one; the dotted line is the magnitude-10 extraction threshold. (B) One cycle, schematically. At issue, both models use earlier quiet intervals; the duration model also uses the length of the completed outburst, here a long one, and places the next onset later. The two lognormal forecasts share one dispersion. The working score is the log probability a forecast assigns to the extracted onset window. (C) Observation years used (upper) and the November 2026 analysis sequence in UTC (lower).

Fig. 2.

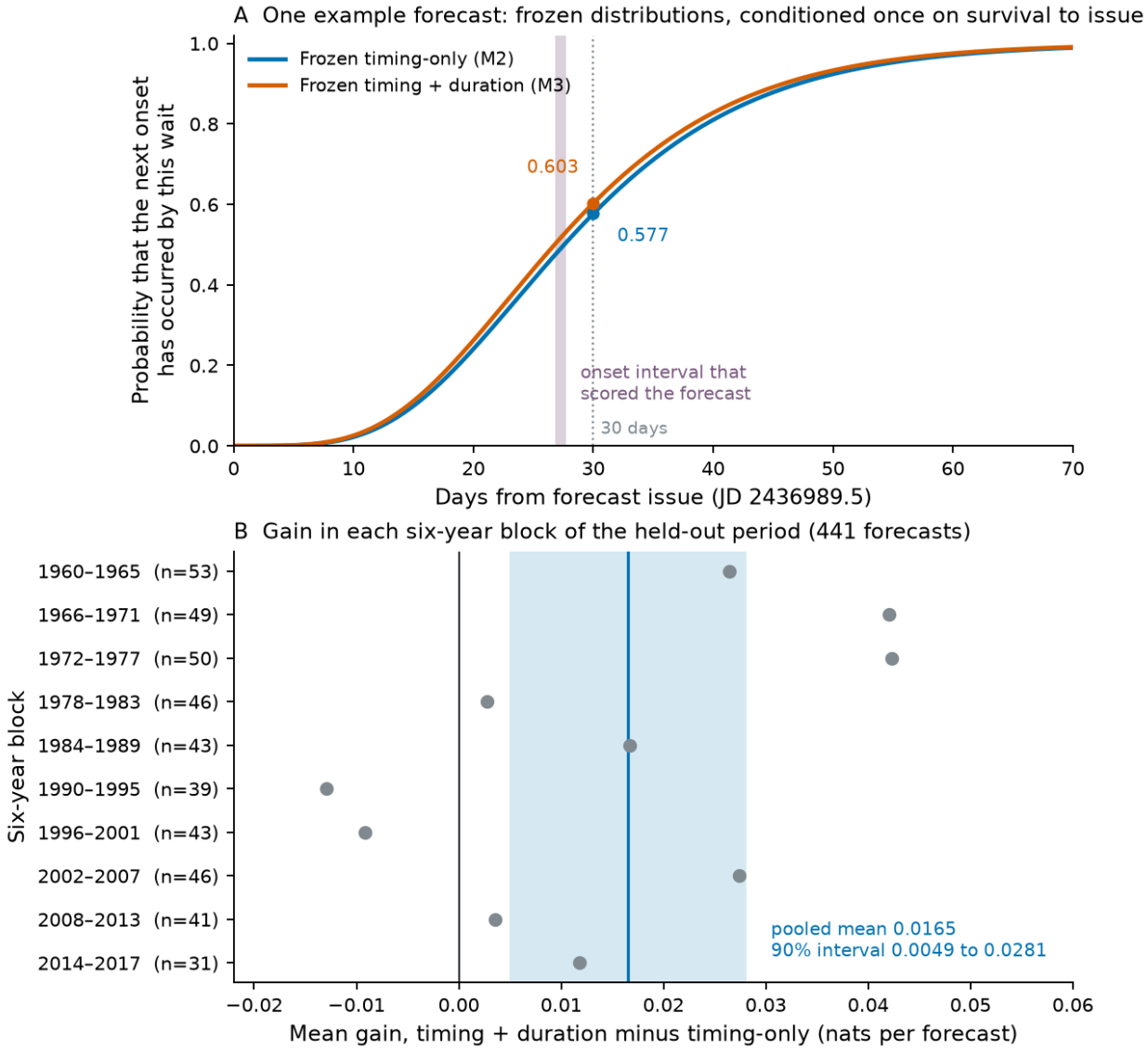


Figure 2: Duration improved the frozen forecasts. (A) The first eligible forecast of the held-out period, chosen by date: probability that the next onset has occurred, by wait since issue and conditioned on survival to issue, under the timing (M2) and timing-plus-duration (M3) models, with the extracted onset window that scored it. M3 placed 0.603 of its probability within 30 days, M2 0.577. (B) Mean paired gain in each six-year block, with the pooled gain of 0.0165 nats per forecast and its 90% interval [0.0049, 0.0281] over 441 forecasts.

426 **Fig. 3.**

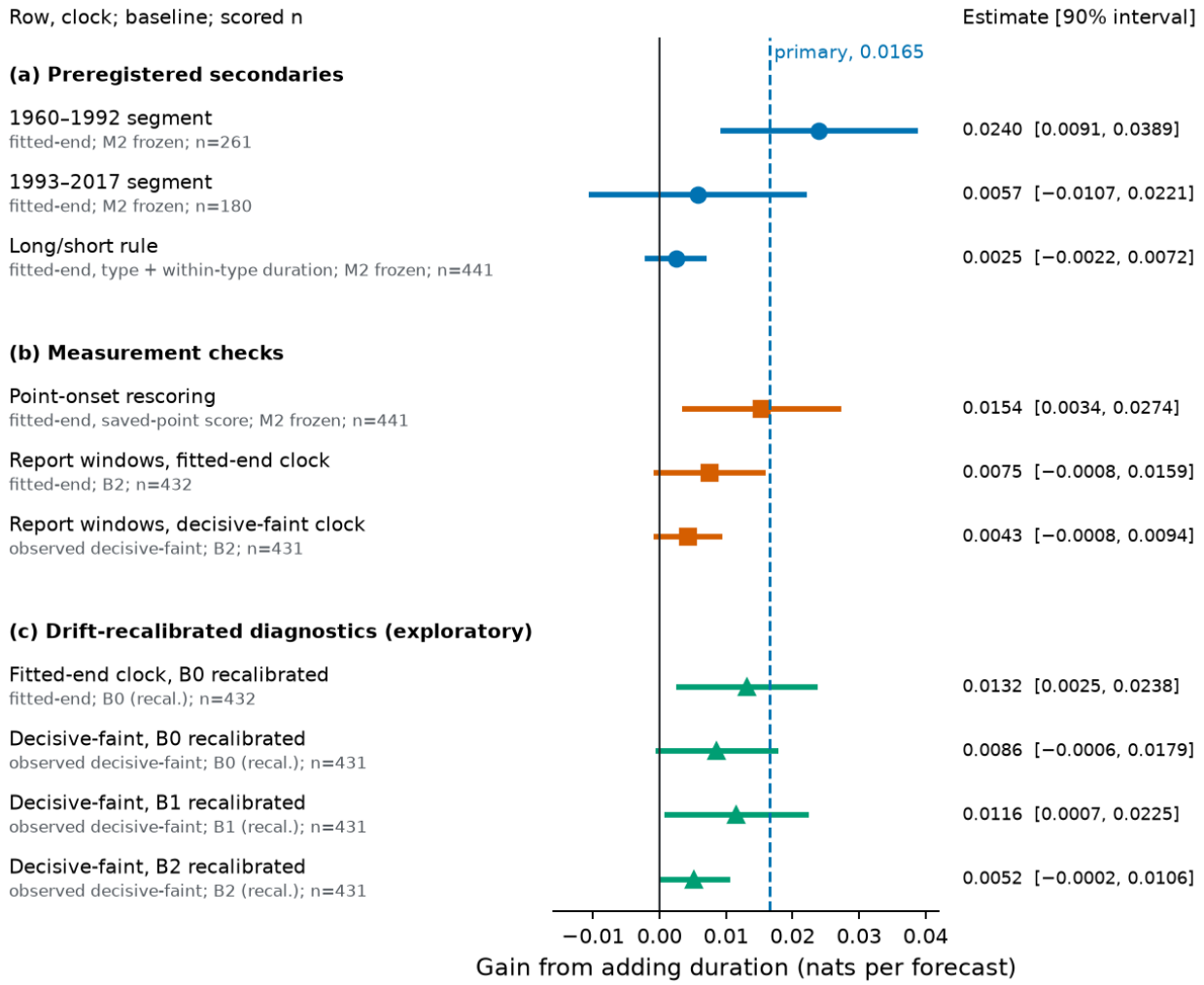


Figure 3: The duration gain under other measurement choices. Selected comparisons, with the primary gain as the dashed line; each row gives its clock, baseline and number of scored forecasts. (a) Preregistered era split and long/short rule, frozen M2 baseline. (b) Point-onset rescoring, and report-window comparisons on the fitted-end (H1) and decisive-faint (H2) clocks with the state-space baseline B2. (c) Drift-recalibrated ("guarded") diagnostics: duration recentred on the preceding ten outbursts and the baseline level updated from preceding outcomes; B0 uses recent-interval covariates, B1 discounted history. Intervals are 90%, from six-year clusters. Groups (b) and (c) reuse opened data.

Fig. 4.

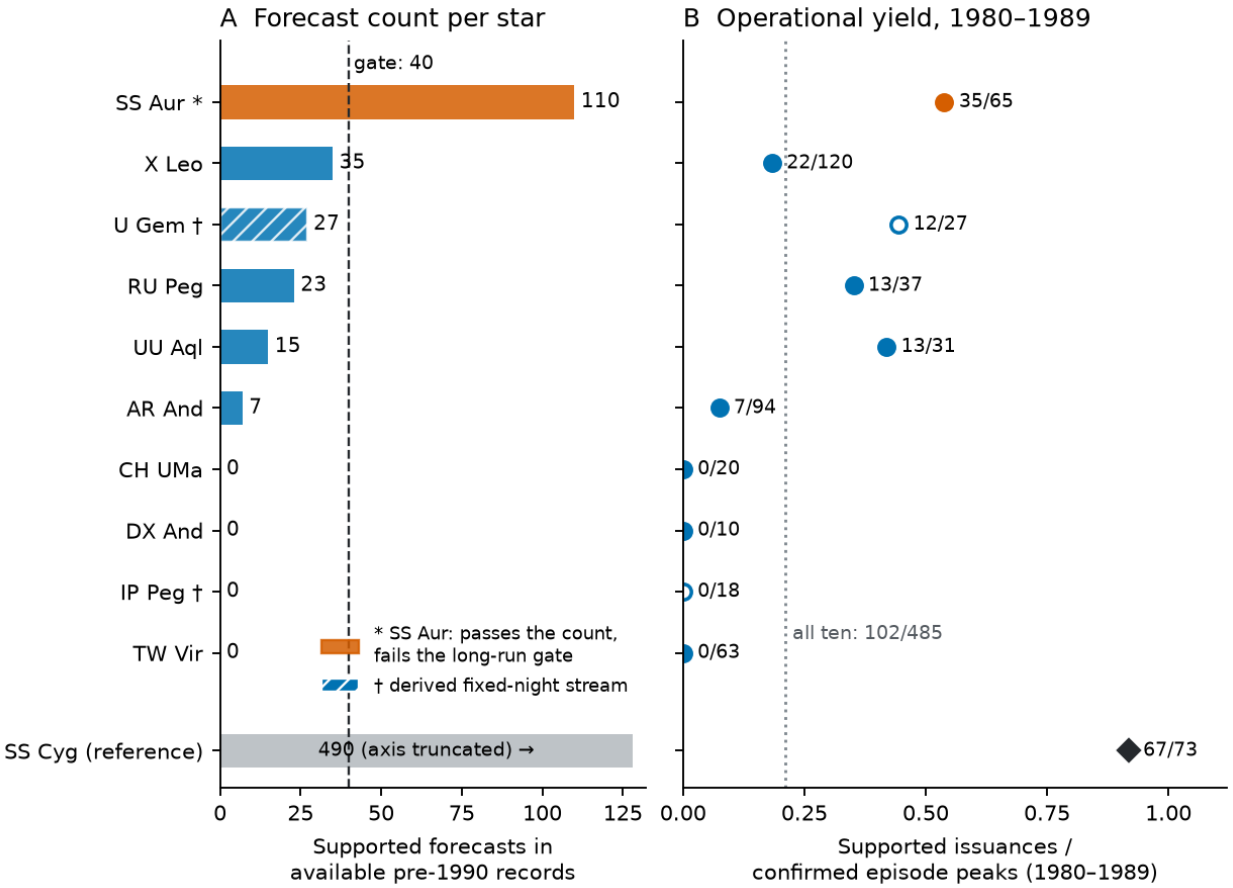


Figure 4: Other dwarf novae's early records support few forecasts. (A) Supported forecasts in each star's available pre-1990 record, against the gate of 40. SS Aur passes the count and fails the long-run check; U Gem and IP Peg use derived fixed-night streams. SS Cyg's count under the same rules is shown for reference (axis truncated); its pre-1990 record includes 1960–1989 of the held-out period, and available spans differ between stars. (B) Supported forecasts divided by confirmed episode peaks, 1980–1989, with counts beside each point; the dotted line pools the ten candidates (102/485) and the diamond is SS Cyg (67/73). Numerators count forecasts by issue date, denominators episodes by peak date. The ratios describe operational yield under the frozen rules.

Fig. 5.

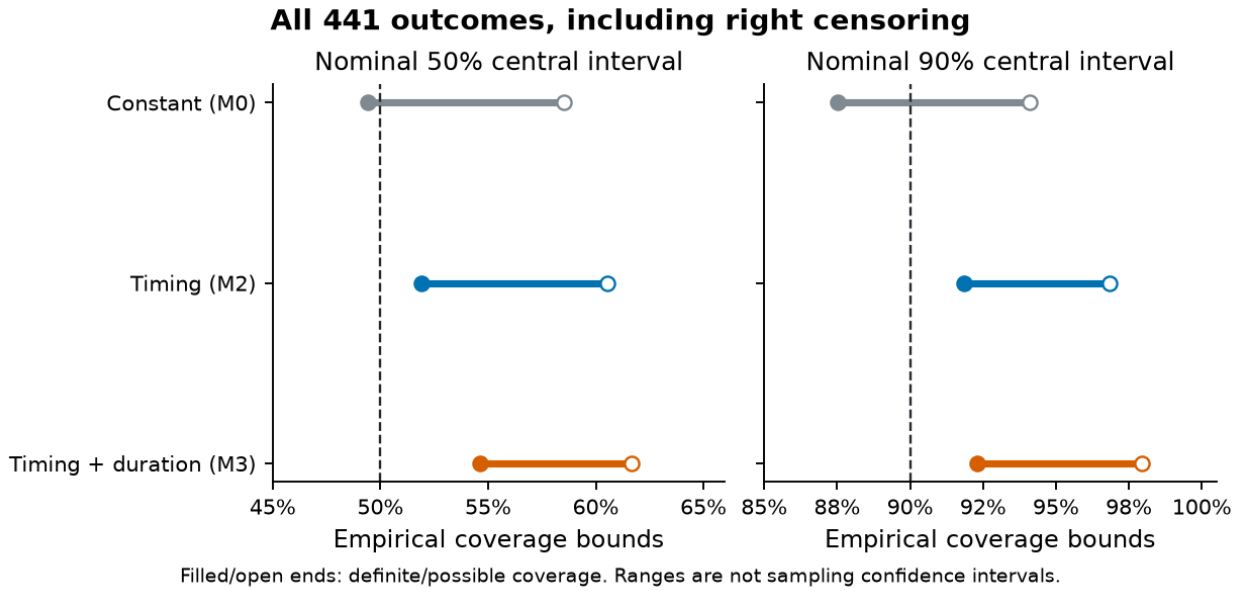


Figure 5: M3's predictive intervals were conservative. Coverage of central predictive intervals for the constant (M0), timing (M2) and timing-plus-duration (M3) forecasts over all scored forecasts, 1960–2017. Each segment runs from definite to possible coverage given the extracted onset windows and censoring; dashed lines mark the nominal fractions.

Fig. 6.

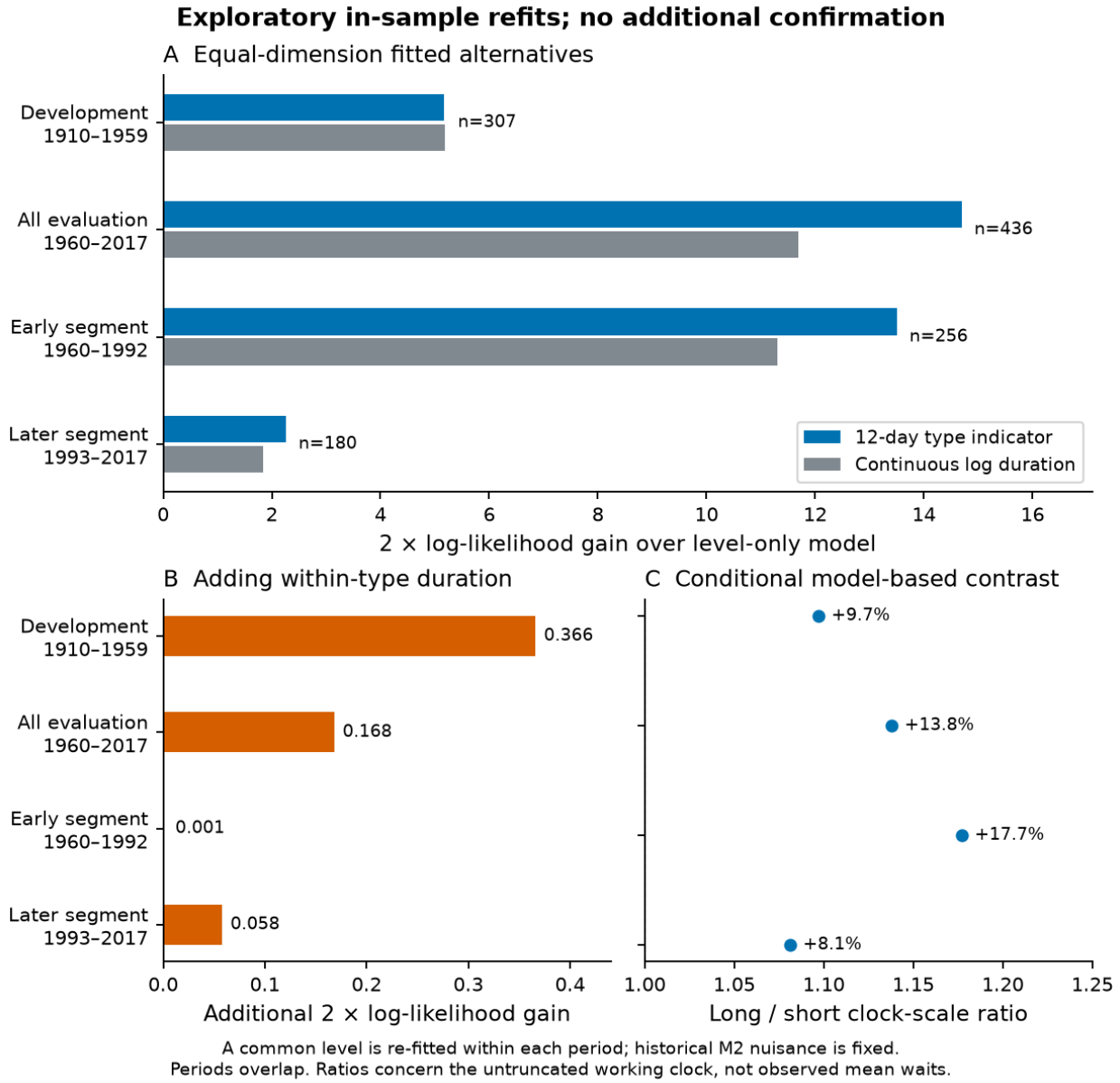


Figure 6: Exploratory long/short comparisons on the evaluation outcomes, with M2's historical parameters fixed and a free common level. (A) Likelihood-ratio statistics for the class indicator and continuous log duration, each against level only; over 1960–2017 they are 14.7 and 11.7. (B) The statistic for adding within-class duration to the class indicator (0.17 over 1960–2017). (C) Long/short ratio of the untruncated working-clock scale, by period.

Fig. 7.

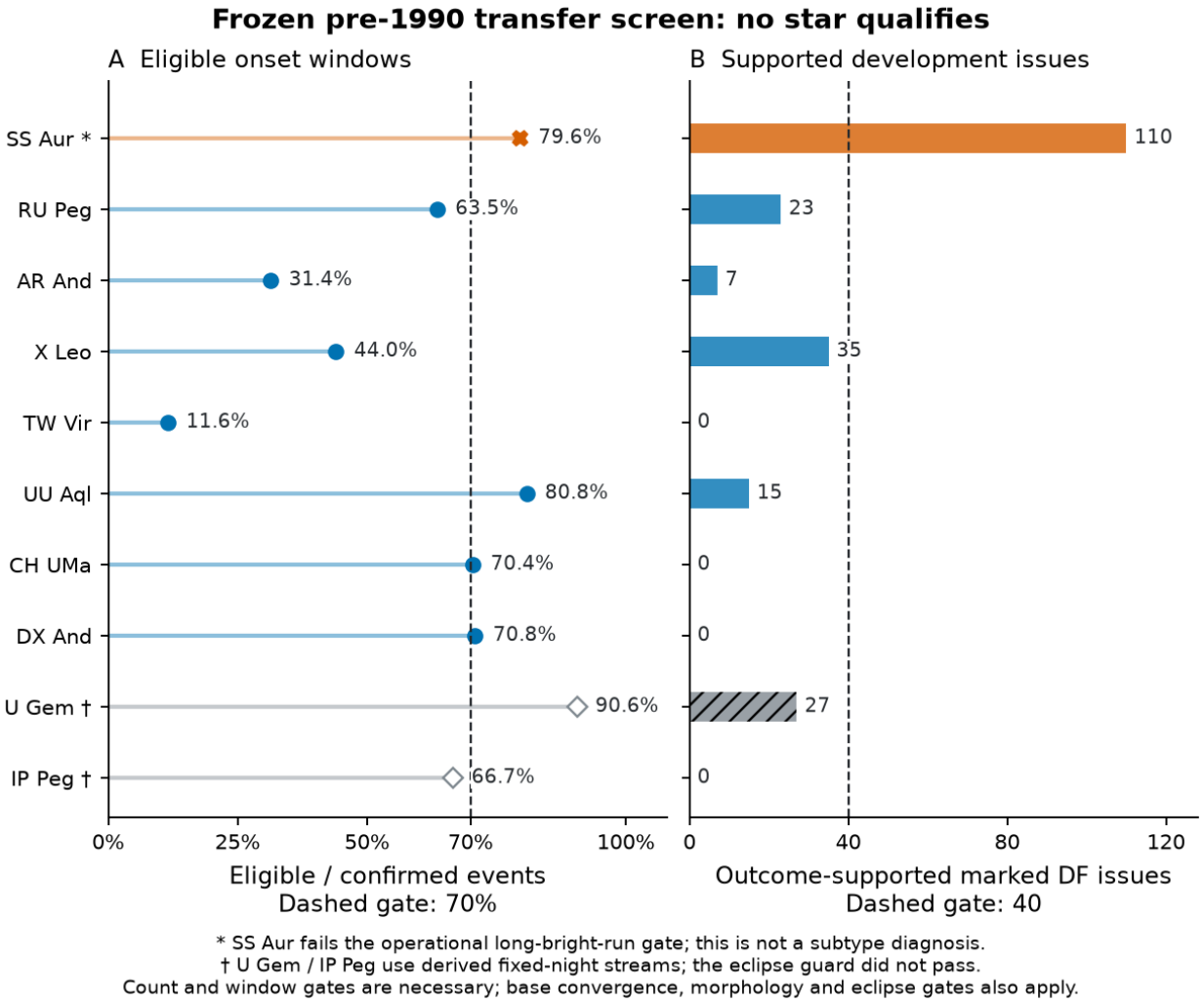


Figure 7: The frozen pre-1990 transfer screen: eligible onset-window fraction against supported forecasts for the ten candidates, with the frozen gates of 70% and 40. SS Aur passes both and fails the long-run check; U Gem and IP Peg use derived night streams.